

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Civil Engineering

Semester : IV

Subject Code & Name: BTCVC403 & Structural Mechanics - I

Max Marks: 60

Date: 24/12/2024

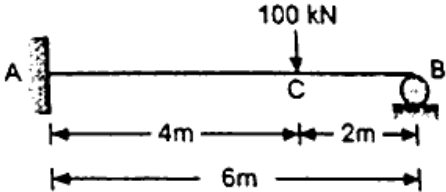
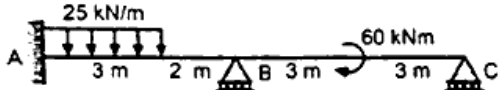
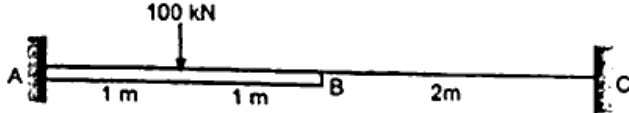
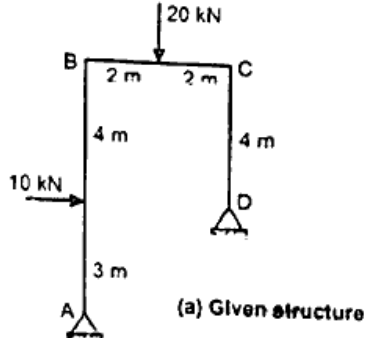
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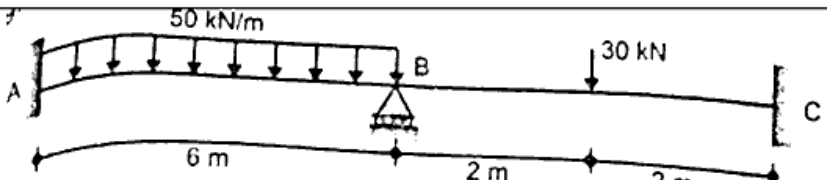
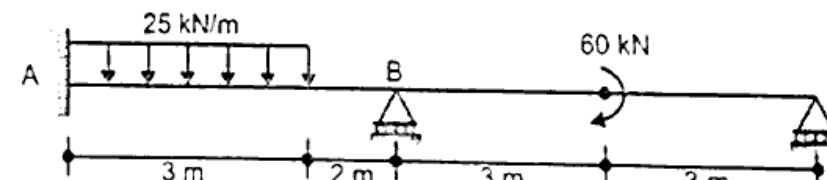
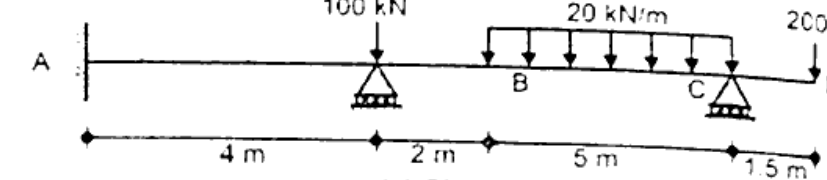
Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Mark s
Q. 1	Objective type questions. (Compulsory Question)		12
1	In cantilever beams, the slope is _____ at fixed end. a. Maximum b. Zero c. Minimum d. Uniform	CO2	1
2	Using Mohr's theorem, calculate the maximum slope of a cantilever beam if the bending moment area diagram is 90kNm ² . Take EI = 4000 kNm ² . a. 0.0225 radian b. 0.0367 radian c. 0.0455 radians d. 0.066 radians	CO2	1
3	In cantilever beam the deflection occurs at _____. a. Free end b. Point of loading c. Through out d. Fixed end	CO2	1
4	Compute the maximum deflection at free end of a cantilever beam subjected to udl for entire span of 1 metres. a. $wl^4/8EI$ b. $wl^4/4EI$ c. $wl^3/8EI$ d. $wl^2/6EI$	CO2	1
5	Calculate the maximum deflection of a cantilever beam with udl on entire span of 3m the intensity of you udl be 25 kN/m. Take EI as 4000 kNm ² . a. 0.052m b. 0.063m c. 0.076m d. 0.09m	CO2	1
6	Which of the following is statically determinate structure? a. Two hinged arch b. Fixed beam c. Double overhanging d. Continuous beam	CO1	1
7	While using three moment equation, a fixed end of a continuous beam is replaced by an additional span of a. Zero length b. Infinite length c. Zero moment of inertia d. None of the above	CO2	1

8	In moment distribution method, the sum of distribution factors of all the members meeting at any joint is always			CO2	1
	a. Zero	b. Less than 1	c. 1		
9	The moment distribution method is best suited for			CO2	1
	a. Indeterminate pin jointed truss	b. Rigid frame	c. Space frames		
10	A beam AB of length 'L' is hinged at its ends and carries a transverse external loading such that the end 'B' is sunk by an amount 'δ'. The fundamental slope deflection equation is			CO2	1
	a. $M_{AB} = \frac{2EI}{L} (\theta_A + 2\theta_B - 3\delta/L) + MF_{AB}$	b. $M_{AB} = \frac{3EI}{L} (2\theta_A + \theta_B - 3\delta/L) - MF_{AB}$	c. $M_{AB} = \frac{3EI}{L} (2\theta_A + \theta_B + 3\delta/L) - MF_{AB}$		
11	Castigliano's first theorem is applicable			CO2	1
	a. A For statically determinate structures only	b. When the system behaves elastically	c. Only when principle of superposition is valid		
12	Clapeyron's theorem is also known as the theory of			CO2	1
	a. 3 - moments	b. 2 - moments	c. Single moment		
Q. 2	Solve the following.				12
A)	List out the different methods to find slope and deflection and explain them in detail?			CO2	6
B)	Derive an expression for slope and deflection for simply supported beam with a UDL throughout span using double integration method?			CO2	6
Q.3	Solve the following.				12
A)	Determine the maximum deflection for the beam as shown below. Assuming $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 1 \times 10^9 \text{ mm}^4$. Using Castiglianos First Theorem.			CO2	6
B)	Determine reaction at B for the propped cantilever AB as shown in fig below. Assuming constant flexural rigidity (EI)			CO2	6

			
Q. 4	Solve Any Two of the following.		12
A)	Determine the fixed end moments for eccentric point load case.	CO2	6
B)	A fixed beam of 2m span is carrying a point load of 50kN at its mid-point. Find the fixed end moment and deflection of the beam at its midpoint.	CO2	6
C)	A beam ABC is simply supported at A and C and is continuous over B. It is loaded with UDL intensity of load on AB is 60kN/m having span 4m and on BC 100kN/m having span 6m. Find moment at B and draw SFD and BMD.	CO2	6
Q.5	Solve Any Two of the following.		12
A)	Analyse the continuous beam ABC as shown in fig below by moment distribution method.	CO2	6
			
B)	Analyse the continuous beam ABC as shown in fig below by moment distribution method.	CO2	6
			
C)	Analyse the continuous beam ABC as shown in fig below by moment distribution method.	CO2	6
	 (a) Given structure		
Q. 6	Solve Any Two of the following.		12
A)	Analyse the continuous beam ABC as shown in fig below by slope deflection method.	CO2	6

	 (a) Given beam		
B)	Analyse the continuous beam ABC as shown in fig below by slope deflection method. 	CO2	6
C)	Analyse the continuous beam ABCD as shown in fig below by slope deflection method. The support B sinks by 15mm. $EI = 3800 \text{ kNm}^2$.  (a) Given beam	CO2	6
*** End ***			